

WEB-BASED ML MODEL EVALUATOR

Functional, Non Functional Requirements and Requirement Analysis

1. Functional and Non-Functional Requirements

1.1 Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale / Comments
FR-01	Functional	Any Analyst can register using the open register process and log in as soon as he completes the registration. No other user, including Admin accounts, is allowed to register by himself. A new Admin account can only be registered by an already created Admin. As soon as a user log in then he is able to use the capabilities offered to him.	High	Pass: a legitimate login gives access to the dashboard for the corresponding role, and an Analyst account is created only by registration; there is no way through which a user can become an Admin through registration. Fail: an illegitimate login gives access anyway, an Analyst accesses an admin-level page, or an Admin role is created by registration.	All else relies on the knowledge of who is signed in. Limiting the way the Admin role can be assigned ensures that no regular Analyst can increase his own access level.
FR-02	Functional	The Analyst uploads a dataset as well as one or more models, where the target variable is specified during the uploading process of the dataset. Each file is checked for its format independently when it is uploaded to the system. When both a dataset and one or more models are uploaded, the system verifies the features of the model against the schema of the dataset, and each verification process should pass before evaluating the corresponding model.	High	Pass: each file is validated individually at upload and each uploaded model is validated to be compatible with the dataset before it can be evaluated; a corrupted file or mismatch with the schema is rejected with a clear reason at the appropriate stage. Fail : a corrupted file is accepted at upload or a model compatibility check is skipped and evaluation proceeds with a model that is not compatible with the dataset.	Because a model and dataset cannot be determined compatible until both files are present, file validation and per-model compatibility are treated as two separate checks, not one combined step. Checking each model separately also corresponds to an Analyst being able to upload several models against the same dataset.
FR-03	Functional	The Analyst selects the task: classification or	High	Pass : The metric and chart options	This follows from DR-03 and DR-04 below:

		regression. The system checks for compatibility between this selection and the model and dataset uploaded before the evaluation can be executed.		do not appear until the declared task has been confirmed as matching both the model and the dataset. Fail: the evaluation may still go ahead even if the stated task does not match the model.	running a metric set inappropriate to a model does not usually result in a runtime error; it returns values that are numerically correct but semantically invalid, so the check has to be done explicitly rather than implicitly by the Analyst.
FR-04	Functional	Once the model and evaluation dataset are confirmed compatible, the system runs the model against the evaluation dataset and calculates the metrics defined for the selected task. For classification, these are accuracy, precision, recall, F1-score and ROC-AUC. For regression, these are MAE, MSE, RMSE and R ² .	High	Pass: predictions run, and computed metric values are correct for an already known test case. Fail: the evaluation finishes and no output, or a metric value, will not agree with expected value.	History and reporting and comparison only carry any weight at all if the previous step is done correctly so these have equally High importance with the other critical assessment steps.
FR-05	Functional	Once an evaluation has been performed, the visualization determined for this task by the system is plotted, a confusion matrix when the task is classification, accompanied by the ROC plot where the predictions are composed of scores/probabilities to achieve ROC-AUC (cf. DR-01), a residual plot in regression case.	Medium	Pass: the visualization corresponds to the metric values calculated for the evaluation, and a ROC curve is generated anytime ROC-AUC was calculable for the model. Fail: visualization missing, mismatched with the type of the task, does not correspond to the actual results, or, ROC curve is shown when there were no prediction probabilities from the model.	Visualizations make evaluation result more approachable and provide direct support to UR-04. This is also consistent with DR-01: ROC curve is not available when ROC-AUC is not available.
FR-06	Functional	The Analyst may choose two or more completed tasks, and compare the metrics side by side.	High	Pass: When the task results for a set of chosen evaluations align perfectly on one view screen. Fail: the results for one	This directly addresses the overall goal of the system; it enables the Analyst to contrast multiple candidate models

				chosen evaluation are non-responsive or different task-types are compared with no warning.	and assess their relative merit based on the presented metrics. It does not actively decide what wins.
FR-07	Functional	Completed evaluations are saved allowing them to be accessed later. A report can be generated by an Analyst based on a single saved evaluation or multiple saved evaluations; it can cover all saved evaluations from which the report has been generated, the model for each and also its dataset. Metrics and visualizations included in the report are always derived from saved evaluation data.	Medium	Pass: An evaluation that has been previously stored and is later re-opened would still retain the same results as were obtained when first performed. A generated report also stores the model, data, metrics and graphs for all evaluations in the report. Fail: If an evaluation is re-opened and the new results vary from what they were previously or the model is not included in the comparison report.	The reports generated from the values from each completed evaluation are to ensure that that reported figure can be saved to view at a later time, without only being able to access it in a specific session.
FR-08	Functional	Admins are able to see all registered accounts. They are able to suspend and unsuspend registered accounts, as well as be able to transfer a registered account between being an Analyst and an Admin or vice-versa. All account-status and account-role transfers are logged within the audit log, noting who made the change and at what time. Administrators are unable to suspend their own account; it is not possible for the last registered account which is an administrator, to be suspended or have its role transferred to that of analyst.	High	Pass: A suspended account may not log on until reactivated. A role change should take place instantly. Logging includes what changed the suspend/demote and by whom and when the change occurred. Request to suspend or demote a single admin role is refused. Fail: A suspended account is capable of logging in. A role change would not be logged. An Admin role is capable of being suspended, or demoted and leaving no admin control.	Admin is a separate actor, thus this is the requirement in which the Admin is actually held accountable for managing the accounts and also managing the roles, with a common safeguard such that the last admin account could never be suspended or demoted (in either cases), thus there will still be system administration accessible.

1.2 Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale / Comments
NFR-01	Non-Functional (Performance)	Up to 50,000 number of rows for the dataset file and 100 MB for model file, evaluation result shall be delivered within 10 sec with one single user load.	Medium	Pass: at least 95% of all tests within the ranges stated above are completed in less than 10s when tests conducted under the stated test load. Fail: tests commonly fail to complete within 10s to the stated test inputs in the specified range.	When an Analyst is analyzing a number of models, each comparison must run back quickly enough so the comparison is still worthwhile. What is too large will now be specified as 50000 rows / 100 MB, this serves to specify what a too large model means when testing NFR-03.
NFR-02	Non-Functional (Security)	Data sets, model objects and results belonging to one account, is inaccessible to another analyst.	High	Pass: If an analyst is not allowed to open another Analyst's datasets, models, or evals - even when they guess a URL and an ID. Fail: Another analysts's data can be accessed via a direct link or on a non-secured endpoint.	Analysts may also upload data sets that they don't want published; this must be done at the account level and not simply concealed within the interface.
NFR-03	Non-Functional (Reliability)	In case of invalid files, inputs not accepted or error during evaluation. The program will always report it with reason.	High	Pass: all the possible failing scenarios tested (file corrupt, schema mismatch, exceed 100MB limit in NFR-01 for the model) return specific and informative error messages, respectively. Fail: one of the above-mentioned scenarios will produce a generic error message (or just no error at all) or will just make the session crash.	Error messages without a description of the failure lead to a greater amount of time needed to diagnose and fix evaluation failures.
NFR-04	Non-Functional (Usability)	The interface shall be so intuitive, that a new user can start uploading a file and reach to the evaluation results screen, without requiring any assistance.	Medium	Pass: A user (who has never seen the flow before) completes the flow unassisted during a usability check. Fail: A user (who has never seen the flow before) gets stuck during some step and does not know what to	The target user needs an immediate response time. Use of external documentation should not be required for routine usage.

				do next.	
NFR-05	Non-Functional (Maintainability / Extensibility)	It should be organized to enable easy later insertion of a new metric, new visualization, or new task type without changing other disconnected components of the system.	Low	Pass: when we're adding a new metric-the metric-calculation module just grows, when we're adding a new visualization-just the visualization module and when we're adding a new task type-just the task-handling logic without changes for upload/storage/authentication. Fail: The modification will always include the modification of upload/storage/authentication code to make sure that the system works correctly.	This is an entire semester-long project so it's not unlikely that future labs are further developing functionality on top of this same pipeline. That is more of a concern with maintainability/extensibility as opposed to scalability in the typical user-based or load-based sense.

2. Requirement Analysis — User, System and Domain

2.1 User Requirements

ID	User Requirement
UR-01	As an Analyst, I want secure registration and login that is clearly separate from Admin access.
UR-02	As an Analyst, I want to upload a dataset and one or more trained models.
UR-03	As an Analyst I want to be given the choice between classification and regression, and told if my selection is incompatible with the file I have uploaded.
UR-04	As an Analyst, I want to understand how well a model performed through metrics and visualizations suited to the task.
UR-05	As an Analyst, I want to compare models and revisit past evaluations without running them again.
UR-06	As an Analyst, I need to be able to export a report for submission or for recording either for one evaluation or for comparison report.
UR-07	As an Admin, I want to view and manage registered accounts, including suspending one if it is being misused.

UR-08	As an Admin, I want to assign and change user roles, with every change recorded for accountability.

2.2 System Requirements

ID	System Requirement
SR-01	Authenticate users, keep track of sessions, and enforce which functionality is available to Analysts versus Admins.
SR-02	Validate each uploaded dataset and model file individually for correct format at the time it is uploaded.
SR-03	After a dataset and one or more model have been defined it must be ensured that each model's required features match the schema of the dataset, and the declared type of the task match what has been provided. Only then may the model be assessed.
SR-04	Run predictions and calculate the metrics defined for the task.
SR-05	Generate the visualizations defined for the task.
SR-06	Store and retrieve evaluation results, scoped to the account that created them.
SR-07	Generate a structured, exportable report covering every evaluation it was generated from.
SR-08	Admin accounts & roles control - suspend, reactivate an account and change the role of a registered account, an Admin role can be granted to a registered user only if an existing Admin allows, an audit log for every account status and role change.

2.3 Domain Requirements

ID	Domain Requirement
DR-01	For classification models, we have measures such as accuracy, precision, recall, F1-score and ROC-AUC. ROC-curve and the confusion matrix (see FR-05) are just plots which give a visual representation of these metrics. For multiclass classification problems, F1-score is averaged using weighted arithmetic average. ROC-AUC cannot be evaluated unless model is capable of returning probabilities or confidence scores from the models. Otherwise it returns that this metric cannot be evaluated for this model.

DR-02	Regression tasks use MAE, MSE, RMSE, and R^2 as metrics.
DR-03	The expected input features of the model are based on the evaluation data schema. Incompatible inputs with a mismatch shall not yield predictions to allow for evaluation but to block it rather than generate the predictions by incompatible inputs.
DR-04	Metrics need to be interpreted according to the declared task; a classification metric shall never appear alongside regression results, or vice versa.
DR-05	Evaluation set and training set must not be identical. The evaluation set must be selected by the Analyst and this choice recorded by the platform without any guessing as to whether it had been used in the training process.